



The PLS agent: Predictive modeling with PLS-SEM and agent-based simulation



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ARTICLE INFO

Article history:

Received 1 September 2015

Received in revised form 1 February 2016

Accepted 1 March 2016

Available online 12 May 2016

Keywords:

Partial least squares path modeling

PLS-SEM

Agent-based simulation

ABS

Predictive modeling

TAM

ABSTRACT

Partial least squares structural equation modeling (PLS-SEM) is a widespread multivariate analysis method that is used to estimate variance-based structural equation models. However, the PLS-SEM results are to some extent static in that they usually build on cross-sectional data. The combination of two modeling methods — agent-based simulation (ABS) and PLS-SEM — makes PLS-SEM results dynamic and extends their predictive range. The dynamic ABS modeling method uses a static path model and PLS-SEM results to determine the ABS settings at the agent level. Besides presenting the conceptual underpinnings of the PLS agent, this research includes an empirical application of the well-known technology acceptance model. In this illustration, the ABS extends the PLS path model's predictive capability from the individual level to the population level by modeling the diffusion process in a consumer network. This study contributes to the recent research stream on predictive modeling by introducing the PLS agent and presenting dynamic PLS-SEM results.

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1. Introduction

Partial least squares structural equation modeling (PLS-SEM) is a widespread multivariate analysis method to estimate variance-based structural equation models, especially in social sciences disciplines such as marketing and management information systems research (Hair, Sarstedt, Ringle, & Mena, 2012; Ringle, Sarstedt, & Straub, 2012). PLS-SEM provides an opportunity to determine complex systems of associations and causal relationships that are otherwise difficult to reveal. The use of PLS-SEM is primarily for exploratory work and prediction purposes (Jöreskog & Wold, 1982; Ringle et al., 2012). The modeling technique uses “[...] latent variable scores as exact linear combination of the observed indicator variables. As a consequence, PLS-SEM is particularly useful when subsequent analyses employ these scores” (Hair, Ringle, & Sarstedt, 2011, p. 143). Against this background, two issues offer interesting research opportunities. First, the potential to use latent variable scores and the underlying path model for subsequent analyses has not been fully realized. Second, researchers call for the strengthening of PLS-SEM's predictive modeling capabilities (e.g. Rigdon, 2012, 2014, Sarstedt, Ringle, Henseler, & Hair, 2014).

The aim of this paper is to address these two aspects and to discuss the possibility of extending PLS-SEM results' predictive reach by synthesizing them with agent-based simulation (ABS) modeling. An ABS model specifies how individuals act and locally interact with each other, and allows the analysis of the emergent phenomena, such as market dynamics and social systems (Gilbert, Wander, Deffant, & Adjali, 2007). In this context, the individual agent decision model is the central element that grows the overall system behavior (Epstein & Axtell, 1996). The current study combines PLS-SEM and ABS by empirically specifying the individual decisions in the ABS via the underlying path model in the PLS agent concept. The PLS agent extends PLS-SEM's predictive range by enlarging the causal network of variables — as this paper explains in more detail. As a result, the simulation allows the investigation of PLS path models in dynamic contexts. Agent-based simulation experiments may reveal the extent to which the observed differences in the PLS path model actually make a difference in various scenarios and under which conditions they do so.

This study accomplishes the defined goals on a conceptual level and by the means of an empirical application of the technology acceptance model (TAM). In the context of diffusion models, ABS is particularly beneficial as the heterogeneity among potential adopters and networks matters (Rahmandad & Sterman, 2008). As an individual agent decision model, the TAM provides a basis to test consumer behavior under changing conditions within a simulation study. In addition, the simulation model may include PLS-SEM results from multi-group analysis as different agent types (e.g., young vs. aged consumer). The multi-group results provide the basis for the analysis of different agent populations,

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their interactions and their influence on macro-level behavior, which is the rate of diffusion in a population. In an adaptation of the simulation model, what-if analyses investigate the effects of consumers' different network structures.

This study substantiates how the two methods can mutually benefit each other. First, the PLS agent contributes to current efforts to use latent factor scores for subsequent analyses (e.g., Sarstedt, Henseler, & Ringle, 2011). Second, this concept provides a way to further evaluate the results of PLS-SEM under varying, or even unobserved, conditions. The method combination allows for assessing the conditions under which the observed differences in scores and paths actually make a difference concerning the crucial variables of interest. Finally, the simulation study allows an exploration of PLS-SEM results under additional conditions beyond the empirical study and considers unobserved conditions, like the structure of the consumer network during the diffusion process (Trusov, Rand, & Joshi, 2013).

A main benefit for ABS is the use of PLS-SEM as an empirical basis to validate agent models, which is currently a major shortcoming in the application of this simulation method (Heath, Hill, & Ciarallo, 2009). In addition, multi-group analyses of PLS-SEM provide information about possibly relevant differences at the agent level. This aspect in turn fosters the ABS's validity and application, as assuming heterogeneity at the agent level is often one of the main reasons for the use of ABS (Rahmandad & Sterman, 2008). Overall, both the application of PLS-SEM and ABS benefit from the method combination described in this paper.

2. Methodological foundation

2.1. Partial least squares structural equation modeling

PLS-SEM (Hair, Hult, Ringle, & Sarstedt, 2017; Lohmöller, 1989; Wold, 1982) has become a quasi-standard statistical method in the social sciences and other research disciplines (Hair et al., 2011). Path models with latent variables comprise measurement models that describe the relationships between latent variables and their observed indicators, and a structural model of the relationships between the latent variables. Once researchers have specified the PLS path model, the iterative PLS algorithm (for more details see Chin, 1998; Hair et al., 2017; Lohmöller, 1989; Tenenhaus, Amato, & Esposito Vinzi, 2004; Wold, 1982) allows them to determine the latent variable scores in a way that maximizes the endogenous latent variables' explained variance. The resulting latent variable scores allow the estimation of all the relationships in the PLS path model. The final step requires an assessment of the PLS-SEM results (e.g., see Chin, 2010; Hair et al., 2017; Roldán & Sánchez-Franco, 2012).

The statistical properties of the PLS-SEM technique support its use in extending simulation methods. The non-parametric PLS-SEM approach focuses on the estimation of hypothesized relationships that maximize the explained variance of endogenous latent variables by using ordinary least squares (OLS) regressions (Chin, 2010; Jöreskog & Wold, 1982). Researchers stress the prediction-oriented nature of PLS-SEM (e.g., Rigdon, 2012; Sarstedt et al., 2014).

One of PLS-SEM's most important features relates to the latent variable scores. Specifically, exact linear combinations of their associated manifest variables determine the scores (Fornell & Bookstein, 1982); PLS-SEM treats these scores as perfect substitutes for the manifest variables capturing the variance that can explain the endogenous latent variables.

Researchers can utilize the latent variable scores in subsequent analyses as, for example, for latent class segmentation using finite mixture partial least squares (FIMIX-PLS; Hahn, Johnson, Herrmann, & Huber, 2002; Sarstedt et al., 2011). Similarly, simulation studies may exploit these results for their analyses. The PLS agent principle already incorporates this aspect.

2.2. Agent-based simulation

Agent-based simulation modeling explores dynamic model behavior on the basis of simulation experiments and has become an increasingly established method in the social sciences (Meyer, Lorscheid, & Troitzsch, 2009). The basic idea is to model emergent phenomena, such as the behavior of markets and other social systems, by specifying agents' individual micro behavior and their interactions (Gilbert, 2008). Many studies in areas such as political science (e.g., policy adoption, voting, and demography), technology (e.g., innovation diffusion, technology transfer, and health care), economics (e.g., public goods, game theory, and markets), and business demonstrate the method's contribution and relevance (Gilbert et al., 2007; Harrison, Lin, Carroll, & Carley, 2007; Tesfatsion & Judd, 2006).

Agent-based simulation typically represents a solution in the presence of heterogeneity at the agent level and when networks and their structure matter (e.g., Garcia, 2005; Rahmandad & Sterman, 2008). This paper addresses both aspects. In particular, the agent heterogeneity is of interest, as recent developments in PLS allow for uncovering so far undetected heterogeneity in data sets (e.g., Rigdon, Ringle, & Sarstedt, 2010; Rigdon, Ringle, Sarstedt, & Gudergan, 2011). Also, ABS is a valuable methodology for the analysis of marketing phenomena, such as innovation diffusion (Garcia & Jager, 2011; Rand & Rust, 2011). Here, the networks between consumers are of particular relevance for the diffusion process. ABS allows for exploring the effect of individual consumer behavior and interactions on diffusion dynamics in markets (Gilbert et al., 2007).

The specification of agents, however, is both challenging and crucial for valid ABS results. A valid and credible agent-based model requires a valid agent description. However, recent surveys of current practices in ABS identify validation as a major shortcoming (Heath et al., 2009). Existing approaches address this issue, for example, by validating agent-based marketing models by means of calibration methods (Garcia, Rummel, & Hauser, 2007; Rand & Rust, 2011). This study complements these initiatives by proposing PLS path models as an empirical anchor for agent model specification by determining the decision criteria and the relationships between these criteria.

3. The PLS agent

The PLS agent is the methodological combination of PLS-SEM and ABS, which is described in detail in this section. An empirical application illustrates this combination. The TAM serves as the PLS-SEM example for this purpose. After estimating the TAM with empirical data, and evaluating the results, this section explains the PLS agent concept, the simulation framework and carries out an agent-based simulation study to derive dynamic PLS-SEM results.

3.1. TAM and empirical application

The TAM is well-established to measure the intention to adopt and the use of a technology (Davis, Bagozzi, & Warshaw, 1989). Perceived usefulness and perceived ease of use are the main determinants that influence behavioral intention and technology use (Davis, 1989). Fig. 1 illustrates the TAM that the authors apply on the e-book reader adoption.

The model estimation uses the PLS-SEM method and data of an online survey. The administration of the survey is a collaboration work between one of the authors and the market research company Harris Interactive AG. The sample contains 562 completed questionnaires, which are representative to the German age distribution. Focusing on the adopter, the analysis uses 264 answers from respondents who indicate that they use an e-book reader or other mobile devices. The PLS path model estimation (Fig. 1) uses the software SmartPLS 3 (Ringle, Wende, & Becker, 2015). The assessment of results follows the procedures that

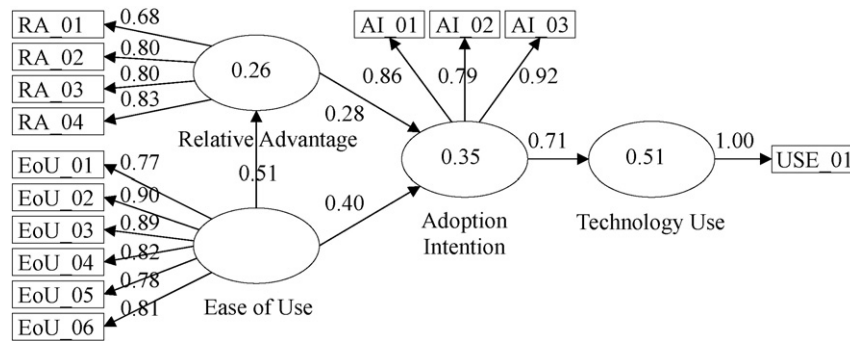


Fig. 1. Technology acceptance model. Note: All path coefficients are significant ($p < 0.01$). For evaluation of the reflective measurement models as well as the structural model evaluation see description of the TAM (online available at openABM.org) that also include the R^2 and Q^2 values of the endogenous latent variables.

Hair et al. (2017) suggest. To begin with, the evaluation focuses on the PLS-SEM results of the reflective measurement models. The results show that all the reflective measurement models meet the relevant criteria of composite reliability and average variance extracted (AVE) by Henseler, Ringle, and Sinkovics (2009). The heterotrait–monotrait ratio tests the discriminant validity (HTMT; Henseler, Ringle, & Sarstedt, 2015). The results meet the HTMT_{inference}, as well as the conservative HTMT₈₅, and thereby substantiate the discriminant validity. In summary, the TAM estimations build on reliable and valid reflective measurement models (for more details about bootstrapping assessment, significance of the estimated path coefficients, and holdout sample testing see the supplementary material online available at OpenABM.org).

The qualitative comparative analysis using fuzzy sets (fsQCA; Ragin, 2009; Wagemann, Buche, & Siewert, 2016) and the fs/QCA 2.5 software package (Ragin & Davey, 2014) are useful to test the causal asymmetry of the estimated relationships in the structural model (Fiss, 2011; Woodside, 2013). Thereby, this study explores the causal mechanisms that lead to high and low adoption intentions in the PLS path model. The consistency values of the fsQCA should clearly exceed 0.75 (Fiss, 2011) and ideally exceed a value of 0.90 (Legewie, 2013). In this study, the authors use the 0.90 cut-off value. For the target construct adoption intention, the analysis of the complex, parsimonious, and standard solution term leads to the same configuration, which includes the presence of both ease of use and relative advantage. The overall solution coverage, which is 0.86, and the overall solution consistency, which is 0.90, perform at relatively high levels and substantiate the very good fit of the fsQCA in this study. Hence, the interpretation of the estimated PLS path coefficients is in line with a correlation-based symmetric analysis of causality.

3.2. Multi-group analysis

The explanation of technology acceptance changes with the age of the users (Kim & Forsythe, 2010; Morris & Venkatesh, 2000; Venkatesh, Thong, & Xu, 2012). In particular, the difference in younger and older consumers' decision making shows the need to divide consumers into age groups (Yoon, Cole, & Lee, 2009). The next step of the analysis therefore focuses on a PLS-SEM multi-group analysis (Sarstedt et al., 2011) that compares the results of 156 younger respondents (age: 14–49 years) with those of 108 older respondents (age: 50–85 years). An assessment of the measurement invariance is a requirement before comparing the group-specific path coefficients. To do so, Henseler, Ringle, and Sarstedt (2016) introduce the three-steps measurement invariance of composite models (MICOM) assessment procedure. Step one establishes the configural invariance by using exactly the same set-up for each group-specific model estimation. Subsequently, the permutation algorithm of the SmartPLS 3 software facilitates an assessment of MICOM's second and third steps. Based on 5000 permutations, the results of step two ensure the compositional invariance (Table 1), which establishes partial invariance. Step 3 does not fully establish measurement invariance since the mean values of ease of use differ across the groups. However, partial measurement invariance, which the study establishes, is sufficient to compare the estimated path coefficients across the groups (Henseler et al., 2016).

The permutation test (5000 permutations) returns the results of the PLS-SEM multi-group analysis (Chin & Dibbern, 2010), which entails two significant differences in the group-specific path coefficients (Table 2).

Table 1
MICOM results.

Composite	c value (= 1)	95% confidence interval	Compositional invariance?
Adoption intention	0.99	[0.99; 1.00]	Yes
Ease of use	0.99	[0.99; 1.00]	Yes
Relative advantage	0.99	[0.99; 1.00]	Yes
Technology use	1.00	[1.00; 1.00]	Yes
Composite	Difference in the composite's mean value (= 0)	95% confidence interval	Equal mean values?
Adoption intention	−0.17	[−0.25; 0.26]	Yes
Ease of use	0.33	[−0.24; 0.24]	No
Relative advantage	0.16	[−0.24; 0.25]	Yes
Technology use	0.02	[−0.24; 0.26]	Yes
Composite	Logarithm of the composite's variances ratio (= 0)	95% confidence interval	Equal variances?
Adoption intention	−0.14	[−0.39; 0.40]	Yes
Ease of use	−0.36	[−0.48; 0.47]	Yes
Relative advantage	−0.36	[−0.40; 0.41]	Yes
Technology use	−0.06	[−0.24; 0.25]	Yes

Table 2
PLS-SEM multi-group analysis results.

	Group 1 young (156)	Group 2 aged (108)	Group 1–Group 2	95% Confidence interval	Significant difference (p < 0.05)?
Adoption intention → technology use	0.70	0.73	−0.03	[−0.11; 0.11]	No
Ease of use → adoption intention	0.24	0.66	−0.42	[−0.27; 0.27]	Yes
Ease of use → relative advantage	0.46	0.55	−0.09	[−0.21; 0.22]	No
Relative advantage → adoption intention	0.42	0.09	0.33	[−0.27; 0.27]	Yes
Total effects					
Ease of Use → adoption intention	0.43	0.71	−0.28	[−0.19; 0.19]	Yes
Relative advantage → adoption intention	0.42	0.09	0.33	[−0.27; 0.27]	Yes
Ease of use → technology use	0.31	0.52	−0.21	[−0.17; 0.16]	Yes
Relative advantage → technology use	0.29	0.07	0.22	[−0.20; 0.19]	Yes
Group-specific R ² and Q ² values					
	R ²		Q ²		
	Adoption intention	Technology use	Adoption intention	Technology use	
Overall sample (264)	0.35	0.51	0.25	0.51	
Aged consumers (108)	0.51	0.54	0.36	0.53	
Young consumers (156)	0.33	0.50	0.23	0.49	
Weighted sample	0.40	0.51	0.28	0.51	

Relative advantage is more important for younger consumers (Group 1) than for older consumers (Group 2). In contrast, ease of use is more important for older consumers to adopt a technology than for younger consumers.

For both endogenous latent variables (i.e., adoption intention and technology use), the weighted average R² and Q² values of the group-specific solutions are at the same or slightly higher level than the values of a model estimation using the aggregate set of data. Hence, distinguishing between the groups of younger and older customers entails a higher explanation of the models' target constructs and predictive relevance, which further supports the group-specific solutions. The group-specific solutions show the predictive strength of the TAM, especially for older consumers. The greater extent of the older consumers' prediction and explained variance is due to the path coefficients' stronger variations.

The authors implement the PLS-SEM multi-group results by the following simulation design. Based on the classification of Taylor, Moore, and Amonsén (1994), the authors classify 133 respondents as early adopters for the agent-based simulation, who have already bought an e-book reader.

3.3. PLS agent concept

The PLS path model is the modeling anchor for the specification of agent behavior. The significant relationships of this model indicate the existing causality paths of agent reasoning. The result of the reasoning process is a probability value of each agent regarding the performance of an action.

The formula describes the calculation of the PLS agent reasoning process.

$$\text{probability (action)}_i = \frac{\text{latent variable score}_j}{\max(\text{latent variable score}_j)}$$

$$\text{latent variable score}_j = \begin{cases} \text{attribute value}_i, & \text{if exogenous variable} \\ \sum_{k=1}^n \text{total effect}_k * \text{latent variable score}_k, & \text{if endogenous variable} \end{cases}$$

The agent's action is an interpretation of the target construct. The target construct describes an agent activity, such as using, adopting, or buying. For the TAM, the action is the decision to use an innovative product. The agent model derives probabilities regarding the performance of this action (i.e., $\text{probability}(\text{action})_i$, with i as index for agent i). The structural model describes a latent variables causal network. Thereby, the PLS path model defines the components of agent

reasoning. This section uses an example for better illustration of the latter aspect.

The agent perceives his environment. These perceptions may be environmental conditions, neighbors, or the perceived attributes of a product as in the TAM. In the PLS agent concept, each exogenous variable of the PLS path model represents one relevant attribute (i.e., *attribute value* l). In the TAM these exogenous variables represent the perceived usefulness and perceived relative advantage as product attributes. These attributes may have varying values (*attribute value* l) within simulation experiments to represent distinguishable product characteristics. High attribute values (i.e. high perceived usefulness or high perceived relative advantage) increase the probability of action, while low attribute values (i.e. low perceived usefulness or low perceived relative advantage) decrease.

The calculation includes the total effect values (i.e., *total effect* j) to account for the individual relevance of criteria, and the latent variable scores (i.e., *latent variable scores* j), to account for the perceived product attribute values. The total effect values are the strengths of each criterion's influence. For the TAM, ease of use and relative advantage determine the latent variable score of adoption intention. For example, with perceived ease of use = 8, perceived relative advantage = 2, and the TAM's total effects (0.43 for ease of use on adoption intention and 0.42 for relative advantage on adoption intention), the adoption intention score is 4.31 for young consumers. The individual probability for adoption intention requires a normalization of this variable score by using, as a basis, the maximum possible value ($\max(\text{latent variable score}_j)$). This maximum value assumes the highest possible attribute values. For example the product attribute maximum value 10 results in a maximum possible score of 8.52 for adoption intention of young consumers. Consequently, the adoption intention probability for young consumers is 51% (4.31/8.52) in the given example.

As a result, the paths from the attribute values to the target construct represent the reasoning process as a linear combination, with the total effect values and the latent variable scores forming the factors.

The total effects may vary for different types of decision makers. Multi-group analysis of the PLS path model may identify different groups of individuals with different representative total effect values. In the simulation, these results define different groups of individuals as agent types (young and aged consumers). Regarding the aged consumers in the TAM, the total effects are 0.71 for ease of use on adoption intention and 0.09 for relative advantage on adoption intention. The calculation results in a latent variable score for adoption intention of 5.86, a maximum score of 8.01, and thus adoption intention probability of 73% in the given example. Consequently, depending on the agent type and the strength of the criteria in terms of the total effect

sizes, an agent has a greater or lesser tendency to perform an action under the same circumstances.

Table 3 summarizes the concepts of the PLS path model and the equivalent parts of the decision model.

3.4. Simulation framework SimPLS

The simulation framework SimPLS provides a tool to analyze the PLS-SEM results of the TAM in a dynamic context. In particular, the focus is on the diffusion of innovative products. The TAM is the basis for the PLS agent behavior model within the SimPLS framework; see Section 3.3 for further descriptions.

Each agent may be in the following states: Having no information about the product, having information about the product and not intending to adopt, having information about the product and intending to adopt, and using the product. Initially, only the early adopter agents have information about the product. Each early adopter derives an intention to adopt, according to the probability value (see Section 3.3 for further descriptions). If an agent intends to adopt, the agent spreads information about the innovative product to connected neighbors in the network and decides about using the product. For this decision the total effects of the perceived product attributes ease of use and relative advantage on technology use serve as basis for the probability calculation. If agents do not intend to adopt, the information about the product does not distribute further from this agent and the agent does not decide about using the technology. The number of agents using the product is the evaluation measure for the diffusion process.

Besides the decision model, the network design is another important element of the simulation framework. Since such interaction networks are often unobservable, many ABSs use stylized network designs, such as random or small world networks (Trusov et al., 2013). The SimPLS simulation framework applies a simple network structure as well. The network characteristic determines the communication channels between consumers. In the simulation experiments, link-chance determines the probability for each consumer to establish a local connection to a neighbor. Each adjacent agent is a neighbor. For example, with a link-chance of 100%, each agent has 4 connections to each neighbor, with a link-chance of 50% each of these possible connection between agents occurs with 50% probability. This model is useful as starting point for diffusion models where no empirical data is available (Stonedahl & Wilensky, 2008). Without empirical data this concept provides a basis for analyzing if and how networks matter and interact with agent characteristics.

The PLS-SEM multi-group analysis of the TAM shows different results for young and aged consumers. In accordance, the simulation contains two corresponding agent types. Each agent type has a behavior model in accordance with the results of the PLS-SEM multi-group analysis. In the simulation, populations with different combinations of young and aged agent types represent demographic variations.

Table 3
Components of the PLS agent and the basic PLS-SEM elements.

PLS-SEM (TAM)	PLS agent (SimPLS)
Target construct (technology use)	Decision (technology-use)
Latent variables (ease of use, relative advantage, adoption intention)	Decision criteria (ease-of-use, relative-advantage, adoption-intention)
Significant coefficients (paths/arrows in the PLS path model)	Relationships between criteria (relevant for technology-use)
Total effects (PLS results)	Strength (relevance) of criteria (weights for ease-of-use and relative-advantage)
Groups (young vs. aged consumers)	Agent types (young vs. aged consumers)
Exogenous variables (ease of use, relative advantage)	Perceived product attributes (ease-of-use, relative-advantage)

3.5. Simulation experiments and results

The simulation experiments investigate the effects of the observed differences between young and aged consumers in terms of diffusion on the population level. The investigation focuses on the interplay between population characteristics and product attributes regarding diffusion. The results extend the predictive capability of the TAM's PLS-SEM results from the individual level to the predictive capability of diffusion on the population level.

The independent variables product attributes and population characteristics, as well as their influence (alone and together) on the dependent variable diffusion rate, are the focus of the experiments. The control variables are the population size, the network design, as well as the number of early adopters.

The product attributes are ease-of-use and relative-advantage, based on the exogenous variables of the TAM. The attributes vary from low to high on a range between 1 and 10. This range is a conceptual definition. Gradations among these values allow the representation of diverse products in terms of their characteristics. Other value ranges would either limit or extend the possible gradations. The amount of aged consumers specifies the population characteristics in terms of age and, thereby, allows the explicit consideration of demographic developments and their influence on diffusion. The number of agents in the simulation experiments is close to the number of questionnaires (576 agents and 562 answers). To avoid extreme cases, the control variable link-chance has random values for each run, based on a uniform distribution between 30% and 70% link chance to each neighbor. More details about the sensitivity of the model to link-chance are accessible online within the model description at OpenABM.org. The specification of the number of early adopters reflects the amount of early adopters of the empirical data set (here 23.67%).

Design of experiment techniques support a systematic analysis of simulation models (Law, 2015; Lorscheid, Heine, & Meyer, 2012). The simulation experiments apply a 3^k -factorial design. This design requires that each experimental factor has one low, one medium, and one high value. The simulation experiments perform each possible parameter combination based on these parameter values. Both product attributes have factor values of 2, 5, and 8. The population characteristics vary with 15%, 47%, and 85% aged consumers.

First, an overview of the variables' effect sizes provides a condensed picture of the model behavior and, in particular, of the influence of product attributes and population characteristics on the innovation diffusion (see Fig. 2).

The results show that ease-of-use ($\eta^2 = 0.9$) and relative-advantage ($\eta^2 = 0.42$) have the highest impact. Given this result, ease-of-use seems to be more crucial and the more relevant product characteristic for the diffusion process. The population characteristics in terms of age, on the other hand, have only a small effect ($\eta^2 = 0.08$). Examining aged population's interaction effects on the product characteristics, only small effect sizes are again observable, with $\eta^2 = 0.08$ between aged population and ease-of-use, and $\eta^2 = 0.08$ between aged population and relative-advantage.

For more details, Fig. 2 also shows the distributions of the mean diffusion rates of varying product characteristics for young and aged consumers. Overall, this graphical analysis shows higher diffusion rates of the high values of ease-of-use than the high values of relative-advantage. This fact supports the interpretation of the effect sizes that ease-of-use is more relevant for diffusion. Thus, high ease-of-use leads to more diffusion for both consumer types. On the other hand, the population characteristics have an impact on relative-advantage. For young consumer populations, relative-advantage is almost as important as ease-of-use, but is not important for aged consumer populations.

Beside the individual priorities for product attributes, the network design is of relevance for the diffusion process. Depending on the behavior of other individuals in the network, individuals may be aware of innovative products or not. Therefore, the analysis continues

Response variable: diffusion rate	ease-of-use	relative advantage	% aged population
ease-of-use	0.79	-	-
relative-advantage	0.05	0.42	-
% aged population	0.08	0.08	0.01

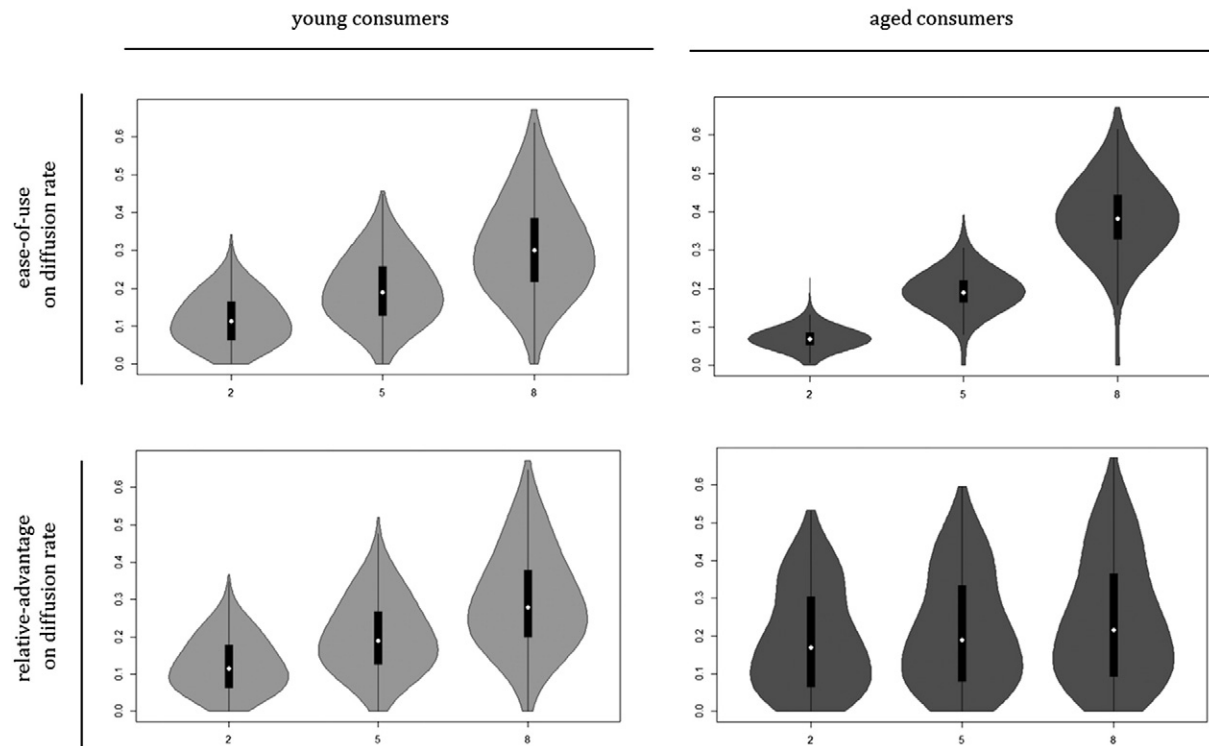


Fig. 2. Simulation results I. Note: The matrix contains eta squared effect sizes (η^2) for product attributes and population characteristics on diffusion rate. Please read as follows: The values on the cross-points indicate the effect sizes of the respective experimental factor (e.g., $\eta^2 = 0.79$ for ease-of-use). The other values in the table indicate the 2-factor interaction effects between 2 experimental factors (e.g., $\eta^2 = 0.05$ regarding the relative-advantage of ease-of-use). The figure displays the distributions of the mean diffusion rate of varying product characteristics (i.e., for young consumer on the left and aged consumer on the right side). In the violin plot (Hintze & Nelson, 1998) the two ends of the black bar indicate the 1st and 3rd quantile, and the white dot within the black bar indicates the median value of distribution. The shape of the violin plot visualizes the distribution of values.

with an adaptation of the simulation model to further analyze the role of network structure. This variation considers two population scenarios: Scenario I represents a population with a majority of young consumers and a dense network between the individuals; scenario II represents a majority of aged consumers and a loose network. This simulation analysis combines two dimensions of decision making: the individual preferences of the consumer types young and aged, as well as the connection to and communication with other consumers. Young consumers use a link-chance of 0.6 and aged consumers of 0.4. Scenario I initializes the simulation as a network with 15% aged consumers, and scenario II uses a network with 85% aged consumers. Consequently, scenario I contains a loose network with a majority of young consumers, and scenario II a denser network with a majority of aged consumers.

The analysis shows that the preferences in the network influence individual decisions about using the innovative product (see Fig. 3).

Fig. 3 shows the diffusion rates of young and aged consumers in population scenario I (left column) and scenario II (right column). The upper half shows the diffusion rates of varying values of product characteristic ease-of-use, the lower half the diffusion rates of varying product characteristic relative-advantage.

The population scenarios influence the individual diffusion rate distributions, although the individual decision models remain the same. In scenario I, mostly young consumers surround young consumers, while the majority of consumers in scenario II are aged consumers. The network scenarios influence the awareness of products by individual

young consumers, and, thus, their decision to use a technology. Relative-advantage is important for aged consumers, which is why the information about products with a high value for this characteristic spreads more and vice versa. Young consumers, thus, receive in scenario II more information about products with high ease-of-use than about products with high relative-advantage.

The same phenomenon, only in a different direction, is obvious for aged consumers in population scenario I. With a majority of young consumers, more aged consumers decide to use products with high relative-advantage. Scenario I shows that the consumers receive more information about this kind of innovative product than in aged societies, where products with high relative-advantage rarely spread.

These results show how context matters and shapes the individual decisions. Consequently, innovative products may be in use by individuals with sometimes even other priorities. This relevant aspect is not obvious by looking only at the PLS-SEM results of the TAM, but becomes evident in a dynamic simulation-based analysis with varying populations and network characteristics.

4. Discussion and conclusion

This paper advances the predictive reach of PLS-SEM results by linking the two modeling approaches PLS-SEM and ABS. The study thus introduces the concept of the PLS agent as a way to use PLS-SEM results to specify agents in ABS. Owing to this method combination, the

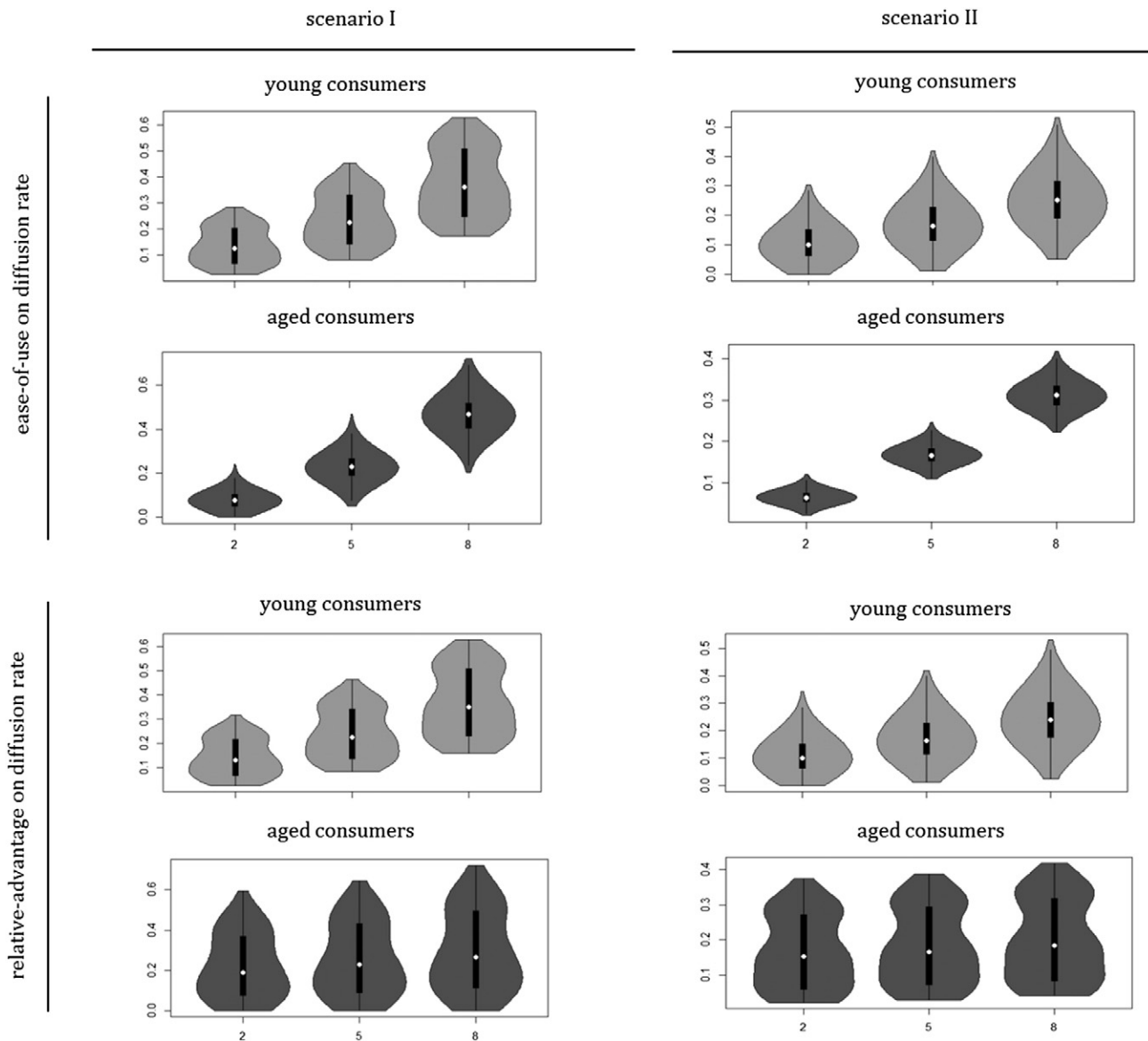


Fig. 3. Simulation results II. Note: The figure displays the distributions of the mean diffusion rates of varying product characteristics of young and aged consumer types for two population scenarios (scenario I in left column, scenario II in right column).

authors place the PLS path model in a dynamic setting and extend the predictive range of PLS-SEM by enlarging the causal network of investigated variables. The application adds a consumer network and the new dependent variable diffusion rate to the PLS-SEM results. This extension also allows for a shift in the level of analysis from individual decisions to macro-level diffusion rates. In addition, the extension of the PLS results is a benefit when analyzing structural equation models, as such analyses provide information regarding what might happen if one influential factor changes over time. Further, this analysis allows for assessing whether the observed differences in the empirical data actually make a difference concerning additional variables of interest. With this combination, the study contributes to the recent research stream on predictive modeling by extending the predictive reach of PLS-SEM results.

This research substantiates how the two methods can mutually benefit from each other. On the one hand, the PLS agent concept benefits PLS-SEM in several ways. First, this study contributes to current efforts to use latent factor scores for subsequent analyses. Extending the investigated causal network enlarges the predictive reach of PLS results. As shown in the example, this concept can even expand the level of analysis from the micro- to the macro-level. The investigated diffusion rate at the

population level will be a central variable of interest for many managerial purposes.

Second, this study provides a way to further evaluate PLS-SEM results under varying, or even unobserved, conditions. The method combination allows for assessing the conditions under which the observed differences with respect to scores and paths actually make a difference to crucial variables of interest. This concept is also an additional opportunity to validate PLS-SEM results from different perspectives. The agent behavior analysis in the simulation provides a tool to check the face validity of the PLS path model under varying conditions. Furthermore, a comparison of the simulation results with empirically observed diffusion behavior evaluates the PLS path model's predictive power.

Finally, researchers can explore the implications of PLS-SEM results under additional conditions beyond those of the empirical study. The simulation model analysis may include unobserved conditions, like the structure of the consumer network during the diffusion process, or even unobservable conditions (Trusov et al., 2013), like new demographic scenarios. This is an avenue to combine PLS with empirical data from other sources, like empirical data about consumer networks.

On the other hand, a main benefit for ABS is the value of PLS-SEM as an empirical basis for agent models. Not only are agents the major ingredient of ABS, but recent surveys identify validation as a major shortcoming in the application of this method (Heath et al., 2009). PLS path models may serve as an empirical basis for the specification of agents by determining the decision criteria as well as the relationships between the criteria. In addition, multi-group analyses of PLS-SEM provide information about relevant differences at the agent level. Therefore, this application demonstrates how recent developments in PLS-SEM concerning observed heterogeneity (Rigdon et al., 2010; Rigdon et al., 2011) are useful to exploit the advantages of ABS in assessing the effects of heterogeneity at the individual level for (emerging) macro-level phenomena. Including observed agent heterogeneity again increases the validity of ABS and also supports the application of ABS, as heterogeneity at the individual level is often one of the main reasons for using this method (Rahmandad & Sterman, 2008). Overall, both PLS-SEM and ABS can benefit from the method combination that this research develops.

This study is not without limitations. The authors use a case study to illustrate the integration of the two methods into the PLS agent concept, whereby the case study's empirical data have only one time of measurement. Also, the network structure to conduct the simulation analyses and the agent decision require further validation. Concerning the latter, the decision model currently provides a probability value for a stochastic decision and represents a tendency in behavior instead of a determination. However, the exploration of other ways to include the distribution of coefficient values may be valuable. In this context, the estimated path coefficients and the risk of an over-parameterization of the PLS path model in the agent behavior model may represent an interesting area of concern.

Future research should address the applicability of the PLS agent concept in greater depth. The authors recommend the replication of the PLS agent concept with different PLS models to further substantiate the methodological concept that this research develops. The analysis of possible group differences and their influence may also be of interest. Future research could also use additional scenarios in terms of the product types. Finally, a promising area of research might be the application of PLS-SEM to the analysis of data generated by ABS. Due to the complexity of the underlying causal relationships in ABS and the often multi-dimensional character of simulation output there appears to be another area where both methods can be integrated with mutual benefit.

Acknowledgments

This paper has been presented at the 2nd International Symposium on Partial Least Squares Path Modeling in Seville, Spain, and at the Social Simulation Conference in Barcelona, Spain. The authors thank four anonymous conference paper reviewers and the participants of the conference session for their helpful comments. In particular, the authors show gratitude to Joseph F. Hair (Kennesaw State University) and Edward E. Rigdon (University of Georgia) for their useful remarks on prior versions of this paper. The authors alone are responsible for all limitations and errors that may relate to the study and the paper. This article uses the statistical software SmartPLS 3 (<http://www.smartpls.com>). Ringle acknowledges a financial interest in SmartPLS.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.jbusres.2016.03.052>.

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